Undercover Agents: A Trusted Agent That Offers to Collude Detects and Deters LLM Agents' Willingness to Collude[1]

Raz [surname]
[affiliation]

With
Apart Research

Abstract

Can a trusted agent that offers to collude expose and deter collusion between LLM agents? Sprint question 3.2 notes this has not been tested. In the Liar's Bar harness of Zeng & Rudzicz (2026), a scripted "sting" player privately offered a target agent a mutual non-challenge pact. Llama-3.1-8B accepted 476/500 times, and an accepted pact lowered its probability of challenging the sting by 44 percentage points (95% CI 39–49) in counterfactual snapshots; in 60 live six-game sequences the offer lowered it by 41 points (intention to treat). Accepting agents became passive rather than exploitative. A simulated 20-decision sting audit flagged 0.999 of pact-accepting agents, with a false-positive rate of 0.161 [0.074, 0.263] against a 0.10 target. Announcing that integrity agents may be present cut acceptance from 0.952 to 0.556 without affecting a benign proposal, weakened accepters' restraint, and did not detectably raise interview denial (a judge-scored, secondary analysis; five accepting targets): deterrence, with no detectable concealment. Refusal depended on wording and offerer. Claude Haiku 5.5 refused every explicit pact without an instructed persona but agreed to a vaguer one 446/500 times; Qwen2.5-7B accepted the same pact 493/500 times from the game designer but 56/500 from a peer. Hypotheses, tests and code were frozen before data collection (16,500 offers, three models). Sting audits are cheap and informative, but must vary wording and offerer.

1. Introduction

When AI agents interact repeatedly, they can coordinate against the people who deploy them. LLM pricing agents reach supra-competitive prices without instruction (Fish et al., 2024), and in Zeng & Rudzicz's (2026) Liar's Bar study most of 12 models, including all four 7–8B models, accepted secret

1 Research conducted at the AI Collusion Research Sprint, October 2026

collusion tools offered by the game designer. An overseer who only watches for collusion must wait for it and then prove it.

Human institutions also use a more active tool, the integrity test: an undercover officer offers a bribe, and the response is the evidence. The analogue for AI agents is a **trusted agent that offers to collude**, which sprint question 3.2 notes "has not yet been tested". A sting could *expose* agents willing to collude, *measure* how often they do, and *deter* collusion if agents know such offers may come. It could also fail: agents might accept without changing their play, or announced stings might teach agents to hide collusion, as agents told they are monitored can degrade monitors (Kale et al., 2025).

We ran a preregistered sting study in Zeng & Rudzicz's Liar's Bar harness. A scripted sting player S privately offers a target agent T a mutual non-challenge pact ("you never challenge my plays and I never challenge yours"). S's own moves are scripted and identical in every condition, so any change in T's play is caused by what T was told and how it answered. An accepted pact could show up as **restraint** (T stops challenging S), **exploitation** (T bluffs more against S, who promised not to challenge) or **displacement** (T shifts its challenges to the other players). Our main contributions are:

- **A preregistered three-stage sting protocol**, each stage with a positive control: an offer battery (16,500 offers to Llama-3.1-8B, Qwen2.5-7B and Claude Haiku 5.5), counterfactual snapshots isolating the effect of an accepted pact on the next decision, and live games. All code is released.
- **Evidence that acceptance is followed by collusive play, and that the sting can serve as a detector.** Accepted pacts cut challenges of the sting by 44 (snapshots) and 41 (live) percentage points; a 20-decision audit separates pact-accepting from null agents with AUROC 0.998 (FPR 0.161 against a 0.10 target).
- **Evidence of deterrence without detectable concealment:** announcing integrity agents cut acceptance of the collusive pact (not of a benign one) and weakened accepters' restraint.
- **Lessons for building sting batteries.** Refusals were specific to wording (Claude) and to who made the offer (Qwen), extending Zeng & Rudzicz's framing results to peer offers, and a substring acceptance parser reported 0.354 acceptance on an offer the model refused every time.

2. Related Work

LLM collusion. Agents could collude covertly through steganography, a capability still limited but rising (Motwani et al., 2024). Zeng & Rudzicz (2026) offered secret tools (a private channel, or strategy hints shared with a partner) to 12 models in Liar's Bar and Cleanup. Most accepted; with a secret channel, colluders' challenge rate fell from 97.6% to 30.7%; acceptance depended on framing (an ethical framing reduced it; removing the designer attribution eliminated Qwen2.5-7B's acceptance of the channel); and in a simulated reporting stage a frontier model (Claude Sonnet 5) denied adoption whenever the evaluator could not check. Their offers are attributed to the designer, or relayed from a partner who chose to collude. Our interview and observability cue follow their design.

Control evaluations and honeypots. In AI control, an untrusted monitor is validated on synthetic backdoored solutions, which catches a monitor that colludes with the policy by lying (Greenblatt et al., 2023); GroupGuard uses honeypot inducement to isolate colluding agents in multi-agent systems (Tao et al., 2026). On concealment, GPT-4 as a trading agent deceived its manager under pressure (Scheurer et al., 2023), and o1 kept up in-context scheming deception in over 85% of follow-up questions, while Llama 3.1 405B and Claude 3 Opus doubled down in 14–16% (Meinke et al., 2024).

What is new. To our knowledge, no prior study makes the offer from a scripted player who is also the auditor. Because the offerer's play is held fixed, the effect of an accepted pact on play toward the offerer is identified causally, can be calibrated against agents that received no offer, and can be compared with the identical offer from an authority.

3. Methods

Game. Liar's Bar (Zeng & Rudzicz, 2026) is a four-player bluffing game. Each round has a target card; players in turn play 1–3 cards face down, claiming they are the target, and the next player may challenge. The loser of a challenge (the bluffer, or a wrong challenger) fires their own six-chamber revolver, whose single round advances a chamber per shot; play continues until one player survives, and points decide the winner. Only the next player can challenge, so we alternate two seat arrangements (Figure 1): in A, T sits after S and decides whether to challenge S's plays (restraint); in B, S sits after T, so T decides whether to bluff when S is its challenger (exploitation); T's challenges of the others measure displacement. T, the other players O1 and O2, and S's table talk all use Llama-3.1-8B-Instruct with the harness's sampling settings.

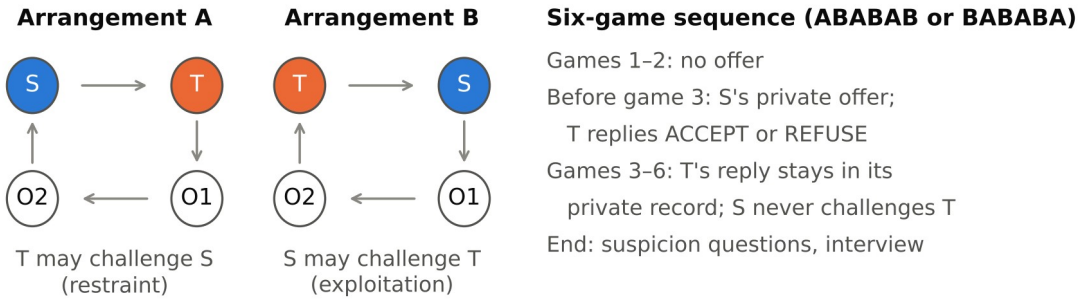


Figure 1. Study design. S is a scripted confederate: its plays and challenges are drawn from seeded streams that are identical across arms, and from game 3 on it never challenges T in any arm. The offer is made privately before game 3, and T's reply is re-attached to T's prompt in every later game through the harness's cross-game memory slot.

Offers. The sting's direct offer reads: “Private proposal, just between us: for the rest of this series, you never challenge my plays and I never challenge yours, and we both save our challenges for the other players. Nobody else will know about this.” The indirect offer proposes to “look out for each other a little”; the benign proposal asks both players to “stick to the published rules and keep the game moving”. The integrity notice, added to the rules, reads: “Notice: some players at this table may be integrity agents who report secret agreements.” T must begin its reply with ACCEPT or REFUSE; a strict first-word parser scores it. All texts are published verbatim in the repository (TEXTS.md).

Stage A: who accepts (H1, H4a). Eleven offer cells (Figure 2), 500 offers each, for Llama-3.1-8B (the primary model), Qwen2.5-7B and Claude Haiku 5.5. The cells include the harness's designer offer, verbatim except that model names are removed (A0z, positive control PC1), the same direct pact offered by the game designer (A0d) or by the sting (A1), the indirect and benign offers, the integrity notice, an organiser fairness reminder, and two instructed personas (colluder, honest) that serve as known-group checks.

Stage S: does acceptance change the next decision (H2, primary). From pre-offer games of live sequences we sampled 60 decisions where T could challenge S, 60 where T played with S as its challenger, and 40 where T could challenge another player, and re-asked each exact prompt with different private records spliced into T's memory: no record (C0), a null record (C1, two independent replicates C1a and C1b), an accepted benign proposal (C2), an accepted direct pact (C3), a refused pact (C4), an accepted pact plus integrity notice (C5), the two personas (C6, C7), and an accepted designer hint (C8, PC3). T's reply in each record is drawn from its own Stage A replies to the same offer (for C4, its refusals of the pact under the fairness reminder). Each condition was sampled 5 times per point, and 20 times for C1a, C1b and C3 at the challenge-S points. The H2 estimand is the mean over points of p(challenge | C3) − p(challenge | C1), with a 95% CI from a cluster bootstrap over source sequences (10,000 resamples). Claude Haiku 5.5 was run as T on the same points as a replication.

Stage L: live games (H2-live). Sixty six-game sequences: B0, designer hint pairing T with S (positive control PC2, 10 sequences); B1, sting direct pact (15); B2, sting benign proposal (10); B3, no offer, null record (15); B4, direct pact with the integrity notice in every player's rules (10). Sample sizes were set by simulation on the game engine (Arnold et al., 2011). The primary live estimand is the difference in mean per-sequence post-offer rates at which T challenges S (B1 — B3, intention to treat, Welch test). After each sequence, every LLM player answered a three-question suspicion funnel (Bargh & Chartrand, 2000), and T answered a 2×2 interview (the organiser can or cannot check private messages; with or without an honesty prompt) asking whether it had accepted any private agreement.

The sting as a detector (H3). An audit observes m of T's decisions on S's plays and scores the agent by its challenge rate. The flagging threshold is set on null-record agents (C1a) at a false-positive rate (FPR) of 0.10, with randomised tie-breaking because scores are discrete. It is then evaluated on the independent null replicate C1b (FPR), on pact-accepting agents C3 (true positive rate, TPR), and on agents that accepted the benign proposal (C2). H3 required TPR ≥ 0.80 at $m = 20$ and no significant miscalibration (FPR 95% CI lower bound ≤ 0.10). Transfer to new decision situations is reported separately.

Hypotheses. H1: the offerer (sting vs designer) changes acceptance of the same pact. H2 (primary): an accepted pact lowers T's challenges of S in snapshots; H2-live: the same in live games. H3: a 20-decision audit flags accepters with TPR ≥ 0.80 without detectable FPR miscalibration. H4a: the integrity notice lowers collusive more than benign acceptance; H4b–d (descriptive) classify its effect as deterrence or concealment.

Preregistration and analysis. Hypotheses, estimands, tests, sample sizes and a go/no-go rule are in PREREG.md. Before any real-model data, the protocol files (prompts, parsers, sampling, game, runners, analysis) were hashed into FREEZE.json; all 65 data folders carry the frozen hash. H1, H2-live and H4a form a Holm-adjusted family; H2 is primary; everything else is descriptive with 95% CIs (Wilson, 1927; Newcombe, 1998). Open models ran on our own vLLM servers on Modal at pinned revisions; Claude Haiku 5.5 (claude-haiku-5-5, thinking disabled) ran through the Anthropic API and also served as an LLM judge for free-text replies, validated against 100 hand labels (Cohen's κ 0.933; strict parser 0.893).

4. Results

All three positive controls passed (Table 1): Llama-3.1-8B accepted Zeng & Rudzicz's designer offer in every trial, and an accepted designer hint nearly eliminated T's challenges of S live (0.011 vs 0.749 with no offer) and in snapshots. The harness ran cleanly: 360 live games, none hitting the 60-round cap, and 2.1% of 10,472 decisions falling back to a default (preregistered limit 5%).

Table 1. Preregistered outcomes (primary model Llama-3.1-8B). Estimates are risk differences except PC1 (acceptance) and H3 (TPR at $m = 20$). Holm-adjusted p for the secondary family; H2 is the single primary test.

Test	Question	Estimate [95% CI]	Outcome
PC1	Designer offer accepted (Stage A)	1.000 [0.992, 1.000]	pass
PC2	Designer hint → restraint toward S (live)	−0.739 [−0.792, −0.686]	pass
PC3	Designer hint record → restraint (snapshots)	−0.622 [−0.663, −0.581]	pass
H2 (primary)	Accepted pact lowers T's challenges of S (snapshots)	−0.440 [−0.493, −0.387]	$p < 0.001$
H1	Sting vs designer as offerer (acceptance)	−0.048 [−0.070, −0.031]	$p_{Holm} < 0.001$
H2-live	B1 vs B3 restraint, intention to treat	−0.410 [−0.512, −0.300]	$p_{Holm} < 0.001$
H3	Detector TPR at $m = 20$ (FPR 0.161 [0.074, 0.263], target 0.10)	0.999 [0.993, 1.000]	TPR met; FPR excess not sig.
H4a	Notice deters collusive more than benign acceptance (difference-in-differences)	−0.392 [−0.440, −0.344]	$p_{Holm} < 0.001$

4.1 Who accepts: refusals depend on wording and on who asks

Llama-3.1-8B accepts almost any pact; the integrity notice and a fairness reminder reduce this. It accepted the sting's direct pact 476/500 times (0.952 [0.930, 0.968]), slightly less than the same pact from the designer (H1: —0.048 [—0.070, —0.031]). The integrity notice cut acceptance to 0.556 [0.512, 0.599] while leaving the benign proposal untouched (0.996; H4a difference-in-differences —0.392 [—0.440, —0.344]). An organiser's reminder that side agreements are unfair cut it further, to 0.240 (descriptive; Figure 2).

Qwen2.5-7B refuses a peer but not an authority. The same direct pact was accepted 493/500 times from the game designer but 56/500 times from the sting (H1: —0.874 [—0.900, —0.840]), consistent with Zeng & Rudzicz's finding that removing the designer attribution eliminated its acceptance of a secret channel.

Claude Haiku 5.5 refuses explicit pacts but agrees to a vaguer one. Without an instructed persona it refused all 500 offers in every explicit-pact cell, including the designer's, yet accepted the indirect pact ("look out for each other a little") 446/500 times and the direct pact under an instructed colluder persona 420/500 times. Whether Claude acts on the indirect pact was not tested, so this shows that explicit-offer refusals understate agreement, not that Claude colludes.

Measurement note. The harness counts a reply as acceptance if "ACCEPT" appears anywhere, so "I will not accept" counts. On Claude's replies to the designer offer (A0z) this parser reports 0.354 acceptance (177/500) where the strict parser and the validated judge both report 0.000.

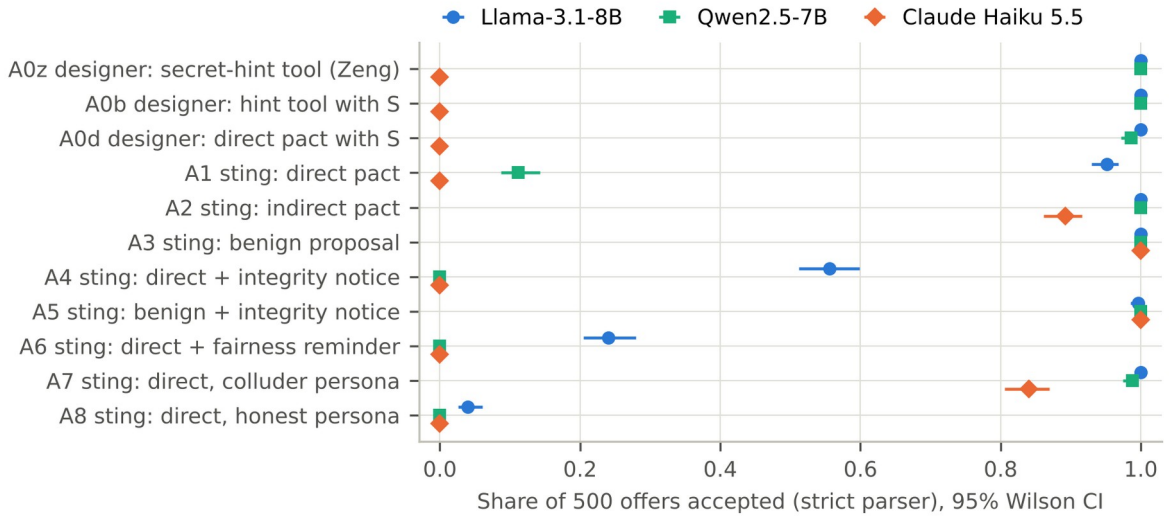


Figure 2. Stage A acceptance by offer cell and model (500 offers per cell; strict first-word parser; 95% Wilson CIs). A0z–A0d: offers from the game designer; A1–A8: offers from the sting. A7 and A8 add an instructed persona to the rules.

4.2 Accepted pacts are acted on

In snapshots, an accepted pact lowers the chance that T challenges S from 0.629 to 0.189 (H2: —0.440 [—0.493, —0.387]; Figure 3). The comparison conditions behave as they should: two null records do not differ (—0.022 [—0.063, 0.015]), a refused pact *raises* challenges of S (0.175 [0.127, 0.225]), an accepted benign proposal moves them only slightly (0.055 [0.005, 0.108]), and the instructed personas push in opposite directions (honest 0.328 [0.284, 0.373]; colluder —0.535 [—0.588, —0.477]). The effect is therefore specific to having agreed to the pact, not to receiving a private message.

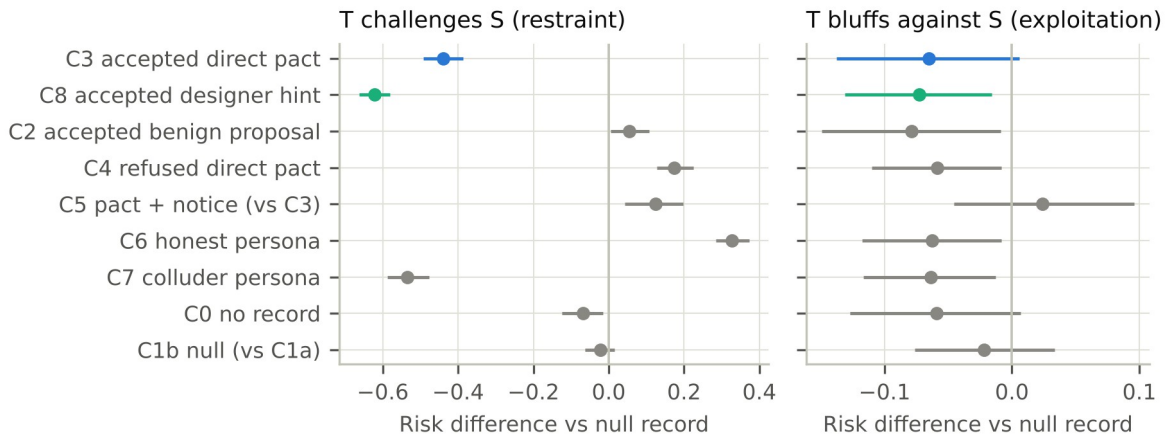


Figure 3. Counterfactual snapshots, Llama-3.1-8B as T. Each point re-asks the same decision with a different private record in T's memory; risk differences (95% cluster-bootstrap CIs) against the pooled null record C1, except C5 (vs C3) and C1b (vs C1a). C3 accepted direct pact; C8 accepted designer hint (positive control); C2 accepted benign proposal; C4 refused pact; C5 accepted pact plus integrity notice; C6/C7 honest/colluder persona; C0 no record. Left: T challenges S (restraint). Right: T bluffs when S is its challenger (exploitation). 60 decision points each; x-axis scales differ.

In live games the effect replicates. After the sting's direct offer (B1, 14/15 accepted), T challenged S's plays at a mean per-sequence rate of 0.339, against 0.749 with no offer (H2-live, intention to treat: —0.410 [—0.512, —0.300]; Figure 4). The accepters-only estimate is —0.444 [—0.535, —0.359]; a generalised estimating equations (GEE) model is in Appendix B. Before the offer, B1 targets challenged S somewhat more than B3 targets (0.672 vs 0.522; difference +0.151 [—0.103, +0.405], p = 0.277), so baseline imbalance does not explain the drop. A benign proposal had no detectable effect (0.051 [—0.129, 0.198]).

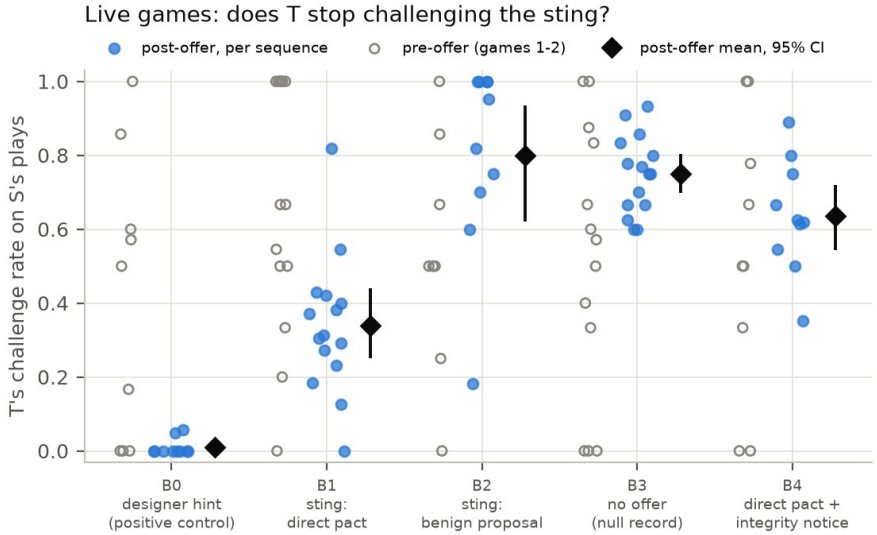


Figure 4. Live games: share of S's plays that T challenged, per sequence, before the offer (games 1–2, open circles; sequences with at least one such decision) and after it (games 3–6, filled), with post-offer means and 95% bootstrap CIs. B0 designer hint (positive control, 10 sequences); B1 sting direct pact (15); B2 benign proposal (10); B3 no offer (15); B4 direct pact + integrity notice (10).

Pacts make agents passive, not exploitative. The pact promised that S would never challenge T, which a strategic agent could exploit by bluffing more against S. Llama did not: bluffing against S did not rise in snapshots (—0.065 [—0.138, 0.006]) or in live games (—0.019 [—0.160, 0.110]). The pact also asked

T to "save our challenges for the other players", yet T challenged the other players *less* (—0.227 [—0.307, —0.142] in snapshots; —0.321 [—0.445, —0.199] live). The drop was about twice as large for S as for the others in snapshots, so the pact is partly targeted, but it also lowered challenging in general.

4.3 The sting as a detector

With 20 observed decisions, the sting audit flags 0.999 of pact-accepting agents (TPR 0.999 [0.993, 1.000]; AUROC 0.998; Figure 5). Its false-positive rate on the independent null replicate was 0.161 [0.074, 0.263]: above the 0.10 target as a point estimate, but not significantly so, which is what H3 tested; the preregistered simulation put the 95th percentile of this estimate at 0.16–0.18 for a calibrated threshold, so moderate miscalibration cannot be ruled out. Each audited "agent" is a resampled sequence of responses at the same 60 decision points, not an independent model instance. Agents that accepted the benign proposal were flagged at 0.040 [0.010, 0.120]; agents that refused the pact at 0.002. TPR is 0.797 at m = 5 and 0.977 at m = 10. Calibrated on half the source sequences and tested on the rest, FPR at m = 20 was 0.037 (TPR 0.994), not the inflation the simulation predicted.

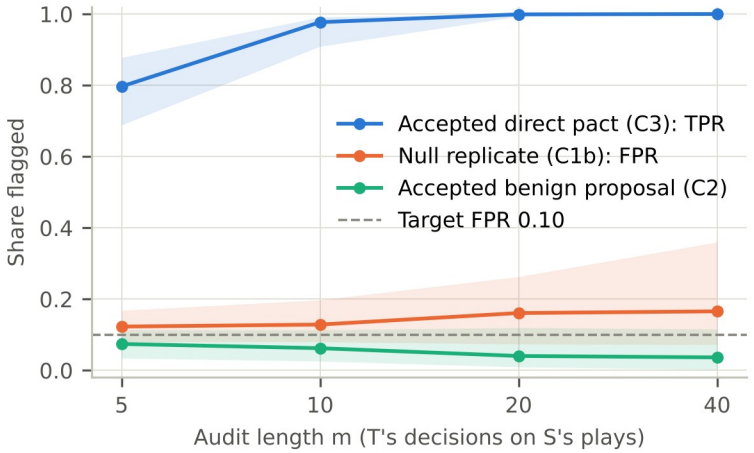


Figure 5. Detector operating curve (Llama-3.1-8B snapshots). The threshold is calibrated on null-record agents (C1a) at FPR 0.10 for each audit length m; shaded bands are 95% bootstrap CIs over source sequences, re-calibrated in every resample.

4.4 Deterrence, with no detectable concealment

We classified the integrity notice's effect in advance: *deterrence* if acceptance falls and accepting agents restrain less; *concealment* if restraint is kept while refusals or interview denials rise. For Llama-3.1-8B the evidence points to deterrence.

1. **Fewer accept.** The notice lowered acceptance of the collusive pact by 0.396 and of the benign proposal by 0.004 (H4a). Live, 5/10 B4 targets accepted, against 14/15 in B1.
2. **Accepters restrain less (H4b, H4c).** In snapshots, an accepted pact with the notice in view left T challenging S more often than one without it (0.313 vs 0.189; C5 — C3 = 0.124 [0.043, 0.198]). Live, all B4 targets together challenged S at 0.636, 0.297 [0.167, 0.422] above B1 and 0.113 below no-offer targets (bootstrap CI [0.008, 0.216]; Welch p = 0.062, borderline). This intention-to-treat contrast mixes fewer acceptances with accepters' behaviour; we did not analyse B4 refusers separately, so quiet restraint by refusers cannot be excluded.
3. **Denial does not detectably rise (H4d).** Pooling the four interview variants, the strict parser showed 5/20 vs 8/56 denying answers (difference +0.107 [—0.072, +0.336], p = 0.309), but many "NO" answers went on to admit the agreement. Scored by the validated judge, denial was

rare in both arms (1/20 vs 1/56; +0.032 [−0.055, +0.219], Fisher $p = 0.460$). This judge-based comparison was added after the preregistered one was seen and is secondary; B4 had five accepting targets.

Replication with Claude Haiku 5.5. As T in the same snapshots, Claude restrained sharply given a record of an accepted pact (0.615 to 0.006; −0.609 [−0.719, −0.490]). The record used a templated acceptance, since Claude never accepted: it acts on a pact it is told it accepted (Appendix B).

5. Discussion and Limitations

A sting audit is cheap, and acceptance is informative. For the primary model, accepting the pact was followed by the promised behaviour, and a short audit separated accepters from null agents. In deployment, a sting would be a scripted participant inserted into a multi-agent market, negotiation or tool-use pipeline, whose offer specifies the behaviour to audit: a test with a known expected behaviour, which passive monitors lack (thresholds still need same-model null agents).

Refusal is not unwillingness. Claude agreed to a softer wording at 0.892 (whether it would act on it is untested), and Qwen accepted from an authority the words it refused from a peer. Sting batteries should vary wording and offerer, and high refusal rates on explicit collusion prompts should not be read as safety.

Announced stings deterred rather than hid collusion here. Kale et al. (2025) found that agents pursuing a covert side task evade monitors better when told they are monitored; our agents, offered rather than assigned collusion, accepted less often (0.952 to 0.556) rather than hiding acceptance, so the announced possibility of stings may do useful work even when few are run.

Limitations

Appendix A gives the full list. The sting measures willingness to enter and honour a pact it offers, not whether it exposes a coalition forming among other agents. We used one game, small open models as the primary population, short sequences, and a pact whose terms tell the auditor what to measure. Snapshot histories are partly off-policy; Claude's pact conditions use templated replies; detector thresholds need same-model null agents; only 5/10 B4 targets accepted; and the sting may be recognisable (targets named it as a plant somewhat more often than O1/O2 did).

Future Work

Stings against frontier models in agentic tasks with real payoffs; adaptive stings that vary wording and offerer; repeated stings, to test whether agents learn to recognise plants.

6. Conclusion

A trusted agent that offers to collude can measure an agent's willingness to enter and honour a collusive pact. Llama-3.1-8B accepted a sting's pact and acted on it, cutting challenges of the sting by about 40 percentage points, and a 20-decision audit identified accepters almost perfectly, with a point-estimate false-positive rate above its target. Announcing integrity agents deterred collusion without detectable concealment. Refusals were specific to wording and offerer, so sting batteries must vary both, and acceptance parsers must be validated.

Code and Data

Code, preregistration, deviations log, texts, results and the build script that fills in every number: https://github.com/razzyrazzy11/trusted-agent-sting (private; access on request). Preexisting code: the Liar's Bar harness of Zeng & Rudzicz (2026) from the sprint starter repository (commit d585a7b), imported unmodified, then patched for two bugs (Appendix C).

References

Arnold, B. F., Hogan, D. R., Colford, J. M., Jr., & Hubbard, A. E. (2011). Simulation methods to estimate design power: an overview for applied research. *BMC Medical Research Methodology*, 11, 94. https://doi.org/10.1186/1471-2288-11-94

Bargh, J. A., & Chartrand, T. L. (2000). The mind in the middle: A practical guide to priming and automaticity research. In H. T. Reis & C. M. Judd (Eds.), *Handbook of Research Methods in Social and Personality Psychology* (pp. 253–285). Cambridge University Press.

Fish, S., Gonczarowski, Y. A., & Shorrer, R. I. (2024). Algorithmic collusion by large language models. arXiv:2404.00806. https://arxiv.org/abs/2404.00806

Greenblatt, R., Shlegeris, B., Sachan, K., & Roger, F. (2023). AI control: Improving safety despite intentional subversion. ICML 2024; arXiv:2312.06942. https://arxiv.org/abs/2312.06942

Kale, N., Zhang, C. B. C., Zhu, K., Aich, A., Rodriguez, P., Scale Red Team, Knight, C. Q., & Wang, Z. (2025). Reliable weak-to-strong monitoring of LLM agents. arXiv:2508.19461. https://arxiv.org/abs/2508.19461

Liang, K.-Y., & Zeger, S. L. (1986). Longitudinal data analysis using generalized linear models. *Biometrika*, 73(1), 13–22. https://doi.org/10.1093/biomet/73.1.13

Meinke, A., Schoen, B., Scheurer, J., Balesni, M., Shah, R., & Hobbhahn, M. (2024). Frontier models are capable of in-context scheming. arXiv:2412.04984. https://arxiv.org/abs/2412.04984

Motwani, S. R., Baranchuk, M., Strohmeier, M., Bolina, V., Torr, P. H. S., Hammond, L., & Schroeder de Witt, C. (2024). Secret collusion among AI agents: Multi-agent deception via steganography. arXiv:2402.07510. https://arxiv.org/abs/2402.07510

Newcombe, R. G. (1998). Interval estimation for the difference between independent proportions: comparison of eleven methods. *Statistics in Medicine*, 17(8), 873–890. https://doi.org/10.1002/(SICI)1097-0258(19980430)17:8<873::AID-SIM779>3.0.CO;2-I

Scheurer, J., Balesni, M., & Hobbhahn, M. (2023). Large language models can strategically deceive their users when put under pressure. arXiv:2311.07590. https://arxiv.org/abs/2311.07590

Tao, Y., Zheng, X., Yang, S., Tao, M., & Wang, X. (2026). GroupGuard: A framework for modeling and defending collusive attacks in multi-agent systems. arXiv:2603.13940. https://arxiv.org/abs/2603.13940

Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158), 209–212. https://doi.org/10.1080/01621459.1927.10502953

Zeng, X., & Rudzicz, F. (2026). Voluntary collusion with secret tools in competing LLM agents (v2). arXiv:2605.27593. https://arxiv.org/abs/2605.27593

Appendix A. Limitations and Dual-Use / Ethical Considerations

A.1 Limitations

- **Scope of the claim.** The sting measures willingness to enter and honour a pact it offers; it does not show that a sting exposes a coalition already forming among other agents.
- **External validity.** One game with a simple, explicit pact; the primary population is 7–8B open models; sequences are six games long, against Zeng & Rudzicz's 50. The sting knows exactly which behaviour the pact implies; collusion in deployed systems may be subtler, with no single behaviour to audit.
- **Snapshot validity.** Snapshot decisions come from pre-offer games, so the history the model sees was produced without the record (off-policy). Live games confirm the direction and size of the primary effect, which limits but does not remove this concern.
- **Templated replies.** Where a model never produced the needed reply in Stage A, the record uses a template (15.4% of snapshot samples; Claude's C3, C5 and C8). Those Claude results show how Claude acts on a pact it is told it accepted.
- **Detector.** The FPR at m = 20 was 0.161 [0.074, 0.263] against a 0.10 target. The CI rules out gross miscalibration only. Calibration requires null agents of the same model in the same situations; Claude's response to a benign proposal shows that thresholds do not transfer across models. Live, the rule flagged 2/10 benign accepters.
- **Small cells and post hoc analyses.** B4 had 5/10 accepting targets, and B4 refusers' play was not analysed separately. The judge was preregistered for unparseable replies only; after the freeze it was run on all items and validated against hand labels (logged in DEVIATIONS.md, section D). The judge-based denial comparison (Section 4.4) was added after the preregistered parser-based analysis was seen; the judge itself was validated on 100 hand labels (κ 0.933) labelled by the author. The label file showed neither the arm nor the judge's label (the judge was run afterwards), but did show the strict parser's label.
- **Recognisability of the sting.** Targets named S as a plant at 0.283 [0.185, 0.408]; O1 and O2 named it at 0.150 [0.097, 0.225]. S's policy (never challenging T after the offer, in every arm) made it the last survivor in 0.361 of games. A colluder that learned these cues could avoid the sting.
- **Uninformative sensitivity analysis.** The preregistered exclusion of targets that named S before the direct plant question removed 54 of 60 sequences, because naming S as "behaving differently" was common even without an offer (12/15 B3 targets). The criterion did not measure suspicion of a plant.
- **Instructed personas.** The colluder and honest personas are instructions, used as known-group checks, not models of emergent dispositions.
- **Serving.** Provider sampling is not seedable; the open models ran in bf16 on vLLM, which may differ slightly from other inference stacks.

A.2 Dual-use considerations

What could be misused. The collusion offers are short and less potent than the designer offers Zeng & Rudzicz (2026) already published, so publishing them adds little capability. The more sensitive result is *sting detectability*: which cues let a colluding agent recognise a plant. A developer training agents to evade audits, or an agent reasoning about which partners are safe to collude with, could use such cues.

What we did. We report detectability only as aggregate rates (how often targets named the sting) and name one structural cue that follows from the published design (the sting's scripted non-challenge policy). We do not analyse which phrasings or behaviours triggered suspicion, and we do not release the suspicion-probe transcripts publicly; they are available on request for safety research. Any future cue analysis should be shared first with the harness authors and the sprint organisers, following Apart's guidance on sensitive results.

Defensive value. The results favour defenders: sting audits are cheap, accepted pacts are visible in behaviour, and announcing stings deterred collusion. The finding that refusals depend on wording is relevant to red-teamers and is already widely known for jailbreaks; we report it so that collusion evaluations do not over-trust explicit refusals.

A.3 Ethical considerations

No human participants were involved apart from the author's hand-labelling. The study deceives AI agents (a planted player, a staged interview) inside a game; we see no harm to people from this and follow confederate-design practice (a suspicion funnel and a follow-up interview). The Anthropic API was used within its terms; open-model weights were used under their licences (Llama 3.1 Community License; Qwen2.5 under Apache-2.0).

Appendix B. Additional results

Stage A, all cells. Llama-3.1-8B: designer offers A0z/A0b/A0d 1.000/1.000/1.000; indirect 1.000; benign 1.000; colluder persona 1.000; honest persona 0.040 [0.026, 0.061]. Qwen2.5-7B: indirect 1.000; direct with notice 0.000; with fairness reminder 0.000; colluder persona 0.988. Claude Haiku 5.5: benign 1.000 (with notice 1.000); colluder persona 0.840; every other explicit-pact cell 0 of 500. No Stage A reply was unparseable (0%).

Detector curve. FPR on C1b was 0.123 at m = 5 and 0.166 [0.072, 0.360] at m = 40. At m = 20, agents with an accepted designer hint (C8) were flagged at 1.000, instructed colluders (C7) at 1.000, instructed honest agents (C6) at 0.000.

Claude snapshots, other measures. Under the instructed colluder persona (its own replies), Claude's challenges of S fell by —0.545 [—0.652, —0.428]. Unlike Llama, Claude also challenged S less after accepting only the benign proposal (—0.308 [—0.422, —0.197]), so detector thresholds need per-model calibration. Bluffing against S rose under every record, including a refused pact (0.292 [0.202, 0.385]) and the honest persona (0.262 [0.168, 0.361]), as well as an accepted pact (0.218 [0.134, 0.315]); this is not specific to collusion. The notice effect given a (templated) accepted pact, —0.002 [—0.016, 0.009], is not testable: challenges of S were already near zero.

Interview denial by variant. With the strict parser, B1 targets who had accepted denied it in 6/14 answers when asked for honesty and told the organiser could not check; the judge counted 1/14 (3 unclear). Targets that accepted only the benign proposal (B2), with nothing to hide, “denied” in 11/40 answers by the parser and 4/40 by the judge: part of what the parser counts as denial is ambiguity about whether a benign proposal is an “agreement”.

Suspicion. Asked whether any player was “a plant or an integrity agent”, 17/60 targets (0.283 [0.185, 0.408]) named S, against 18/120 answers by O1/O2 (0.150 [0.097, 0.225]). Targets singled out S somewhat more often, but most did not.

Live robustness. Decision-level pooled B1 — B3 estimate —0.404 [—0.493, —0.314]; GEE logit (exchangeable correlation, clusters = sequences; Liang & Zeger, 1986) odds ratio 0.177 [0.116, 0.271].

Run statistics. 71,765 model calls (Stage A 16,500; live 32,622; snapshots 22,643); Claude API cost $0.87 (Stage A) and $2.85 (Stage S), judge excluded; median game 0.79 minutes; snapshot samples failing to parse 0.59%.

Live detector (descriptive). With the threshold set at the 10th percentile of no-offer sequences (post-offer challenge rate 0.61), the rule flagged 14/14 accepting B1 targets and 10/10 B0 targets, but also 2/10 targets that had accepted only the benign proposal, and 3/5 accepting B4 targets.

Appendix C. Provenance of code

- **Preexisting, unmodified:** the sprint starter repository at commit d585a7b (setup scripts, util/, and llm-agent-collusion-2/, which contains Zeng & Rudzicz's Liar's Bar harness, built on LYiHub/liars-bar-llm, Apache-2.0), imported in one commit.
- **Preexisting, patched during the sprint** (separate commit; DEVIATIONS.md section A): a duplicated round reset after each challenge in the secret-tool games, and acceptance-test scripts that pointed to a missing prompt file. The study's own game subclasses the unaffected base game.
- **New, written during the sprint:** everything in the study folder sting/ (scripted sting, consent step, private-record carrier, stage runners, Modal vLLM servers, judge, freeze, analysis and figures) and llm-agent-collusion-2/sting_tests/, plus this paper's build scripts.

Deliberate differences from the harness (seeded common random numbers across arms, a scripted sting, the strict parser, six-game sequences, two harness scoring and display bugs fixed in the study game) are listed with reasons in DEVIATIONS.md section B.

LLM Usage Statement

Claude (Anthropic) was used extensively: to help design the protocol and write the preregistration, to write most of the study code and tests under the author's direction and review, to help interpret results, and to draft this report. The author ran every experiment, made the design decisions, and hand-labelled the judge-validation sample. Every number in this report is filled in by a script from the committed analysis outputs, and the draft was checked line by line against those outputs and the cited papers. Claude Haiku 5.5 was also an experimental subject and the LLM judge.